

# Chemical energy

There is energy stored inside all substances in their chemical bonds. These are the forces that hold atoms together. The chemical name for water is  $\text{H}_2\text{O}$  because it is made of two hydrogen atoms and one oxygen atom, joined together with chemical bonds. Energy is released when new bonds are made during chemical reactions.



Potassium iodide is a catalyst.

Mix of hydrogen peroxide, food colouring, and washing-up liquid.



The washing-up liquid and oxygen bubbles make foam.

## Elephant's toothpaste

In this reaction, a chemical called hydrogen peroxide breaks down into water and oxygen. This happens naturally, but potassium iodide is added as a catalyst. Catalysts make chemical reactions happen faster.

## Chemical reaction

The hydrogen peroxide breaks apart to form oxygen and water. The oxygen gas bubbles out into the washing-up liquid to make foam. The formation of oxygen gas also produces heat.